ECON 423/523-Lab-7

Sonan Memon

University of Oregon



Experimental Economics

- The so-called credibility revolution meant that economics moved toward experiments, randomized controlled trials and the like.
- This is not the same as using instruments with observational data which was the classical approach.
- Such identification is potentially superior to methods using observational data from for instance, surveys.

Randomized Controlled Trial (RCT)

Example: Does a job-training program increase workers' wages?

Suppose we have 200 workers randomly assigned to two groups:

- **Treatment Group:** 100 workers receive job training.
- **Control Group:** 100 workers do not receive training.

After the experiment, average monthly wages are:

Group	Average Wage
Treatment	\$3,500
Control	\$3,000
Difference	**\$500**

The estimated Average Treatment Effect (ATE) is:

$$\widehat{ATE} = \bar{Y}_1 - \bar{Y}_0 = 3500 - 3000 = \$500$$

Potential Outcomes Framework

For each individual i, define:

- $Y_i(1)$: Potential wage if the worker receives training.
- $Y_i(0)$: Potential wage if the worker does not receive training.
- $D_i \in \{0, 1\}$: Treatment indicator.

The individual treatment effect is:

$$\tau_i = Y_i(1) - Y_i(0)$$

The Average Treatment Effect (ATE) is:

$$ATE = E[Y_i(1) - Y_i(0)]$$

Fundamental Problem of Causal Inference:

We can observe only one potential outcome for each individual.

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

Conditional Independence Assumption (CIA)

Let X_i denote observed pre-treatment characteristics such as education and experience.

The **Conditional Independence Assumption** states:

$$\boxed{(Y_i(1), Y_i(0)) \perp D_i \mid X_i}$$

Interpretation:

- Conditional on X_i, treatment assignment is independent of potential outcomes.
- After controlling for X_i, treated and untreated individuals are comparable in terms of their potential outcomes.
- CIA is also called **selection on observables** or **unconfoundedness**.

CIA in Randomized Experiments

In an RCT, treatment is randomly assigned.

Therefore, randomization guarantees:

$$\boxed{(Y_i(1), Y_i(0)) \perp D_i}$$

This is **unconditional independence**, a stronger condition than CIA.

Consequently, under standard assumptions:

$$\begin{aligned} ATE &= E[Y_i(1)] - E[Y_i(0)] \\ &= E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] \end{aligned}$$

Key Insight: Random assignment eliminates systematic selection bias in expectation.

CIA in Observational Studies

Suppose workers voluntarily enroll in job training.

- More motivated workers may be more likely to enroll.
- Motivation may also independently increase wages.
- The treatment and control groups may therefore differ systematically.

Without controlling for confounders:

$$(Y_i(1), Y_i(0)) \not\perp D_i$$

However, if all relevant confounders are captured by X_i, we may assume:

$$\boxed{(Y_i(1), Y_i(0)) \perp D_i \mid X_i}$$

Important: Unlike randomization, CIA in observational studies is an identifying assumption, not a guarantee.

Treatment with OLS

- $\epsilon_i \sim N(0, \sigma_i^2)$
- $\sigma_i^2 = \sigma_1^2,\ \forall i \in [0, N_1]$ and $\sigma_i^2 = \sigma_0^2,\ \forall i \in (N_1, N_2]$.

$$Y_i = \alpha + \beta T_i + \epsilon_i$$

$\alpha = E(Y_i|T_i = 0)$.

$\beta = E(Y_i|T_i = 1) - E(Y_i|T_i = 0)$.

Thus, OLS estimator is:

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0$$

Variance:

The variance of the sample mean for the treatment group is:

$$
\begin{aligned}
Var(\bar{Y}_1) &= Var\left(\frac{1}{N_1}\sum_{i=1}^{N_1} Y_i\right) \\
&= \frac{1}{N_1^2} Var\left(\sum_{i=1}^{N_1} Y_i\right) \\
&= \frac{1}{N_1^2}(N_1\sigma_1^2) \\
&= \frac{\sigma_1^2}{pN}.
\end{aligned}
$$

Variance:

Similarly, for the control group:

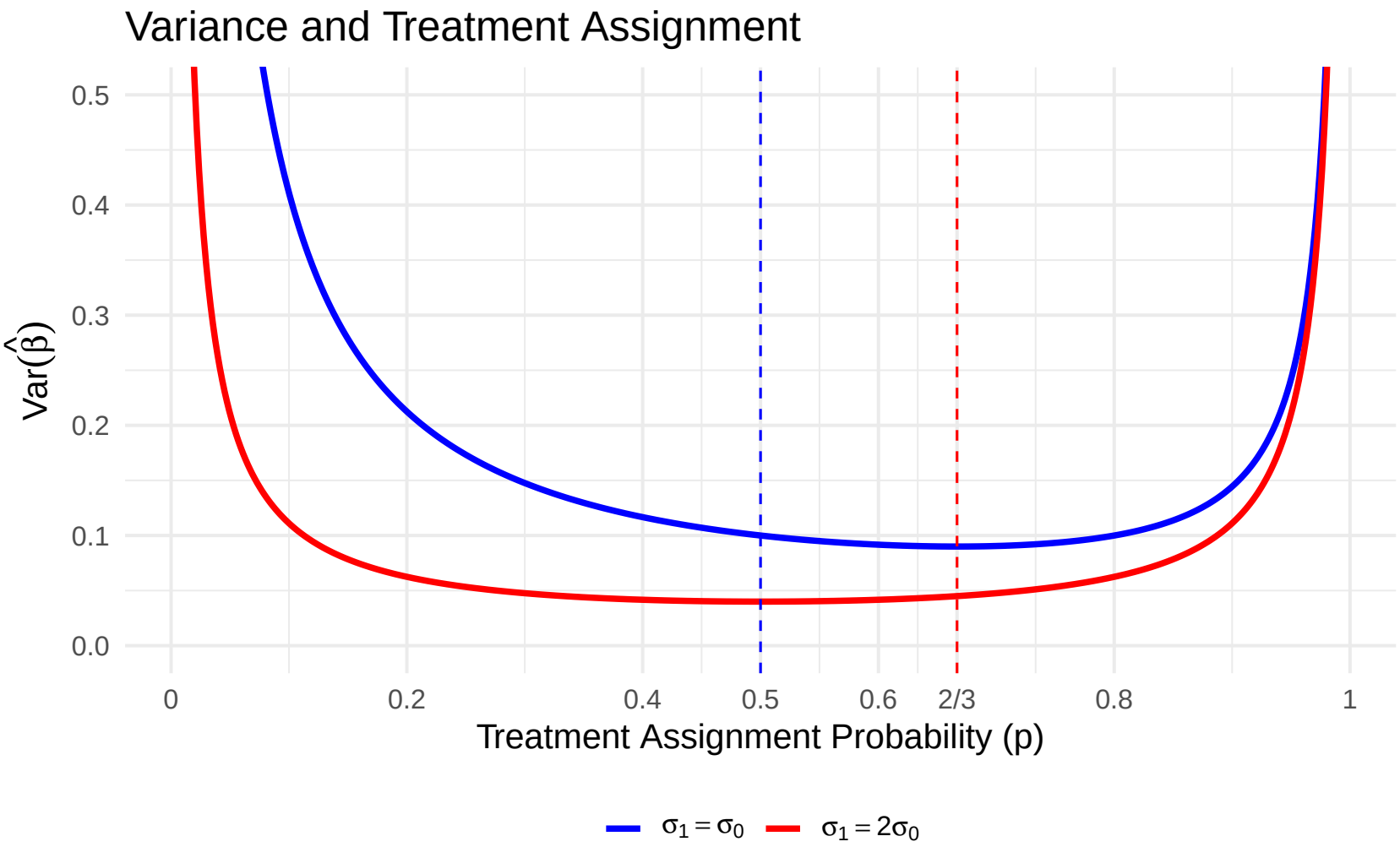
$$Var(\bar{Y}_0) = \frac{\sigma_0^2}{(1-p)N}.$$

Therefore, assuming independence between the two groups:

$$Var(\hat{\beta}) = \frac{\sigma_1^2}{pN} + \frac{\sigma_0^2}{(1-p)N}$$

$$\boxed{Var(\hat{\beta}) = \frac{(1-p) \times \sigma_1^2 + p \times \sigma_0^2}{p \times (1-p) \times N}}$$

Variance Minimization



Minimum Detectable Effect

$$MDE = (t_{0.975} + t_{0.80}) \times SE(\hat{\beta})$$

With equal variance across treatment and control groups,

$$MDE \approx 2.8 \times \sqrt{\frac{\sigma^2}{p \times (1-p) \times N}}$$

- For instance, if $N = 500$, $p = 0.5$ and $\sigma^2 = 1$, $MDE \approx 0.250$.

Power of Deteching β:

- $$\begin{aligned} P(\hat{\beta} > 1.96 \times SE(\hat{\beta}) \mid \beta = 0.24) &= P\left(\frac{\hat{\beta} - 0.24}{SE(\hat{\beta})} > 1.96 - \frac{0.24}{SE(\hat{\beta})}\right) \\ &= P\left(Z > 1.96 - \frac{0.24}{SE(\hat{\beta})}\right) \\ &= P(Z > -0.723) \\ &= 1 - \Phi(-0.723) \approx 0.765. \end{aligned}$$

- Thus, $\beta = 0.24$ is not detectable with power of 80% unless we increase N or assume lower σ^2.
- With $N = 1000$, $MDE \approx 0.1771$, so $\beta = 0.24$ is detectable with power greater than 80%.

Minimum Detectable Effect

I assume SE $= 1$ for graph below.

